

Data Model

Entity-relationship diagrams, foreign keys and feature-by-feature table maps for the postgraduate research lifecycle platform.

66	103	20	1
TABLES	FOREIGN KEYS	FEATURE MODULES	POSTGRESQL DATABASE

Generated directly from the application schema (SQLAlchemy metadata) — every table, column marker and relationship shown here is read from the code, not drawn by hand.
PostgreSQL 16+ · Migration head f6_assistant_notif · 2026-08-24

Diagram notation

Element	Meaning
Coloured header	A table. The colour is its feature area (see below).
PK	Primary key — every table uses a UUID id.
FK	Foreign key — the column points at another table's id.
Arrow	Points from the <i>child</i> (the table holding the FK) to the <i>parent</i> it references. Parents sit above their children.
Dashed box	A table owned by another feature, shown only for context (marked <i>ref</i>). Its full diagram lives in its own section.
+N more	The box shows keys and notable columns; N further columns exist. Full counts are in the inventory (§3).

Conventions that shape every table

Surrogate keys everywhere. Every table has a UUID id. Natural identifiers (student_ref, award_ref, code) are unique columns, never the primary key — so a mis-keyed reference can be corrected without breaking relationships.

History is preserved, not overwritten. Where a fact changes over time — supervision, funding, a person's identities — the table carries valid_from / valid_to. A change closes one row and opens another. This is why the platform can answer "what did we know on the day the decision was taken".

Deletion is rare and deliberate. Domain records transition or close; they are not deleted. Hard deletes exist for contact details, uploaded documents, draft report mappings and unused reference values only — see §7.

Feature areas

-  **People & Access** — 2 modules, 11 tables
-  **Recruitment & Admissions** — 2 modules, 9 tables
-  **Student Journey** — 3 modules, 13 tables
-  **Funding** — 1 modules, 4 tables
-  **Thesis & Completion** — 2 modules, 6 tables
-  **Research & Intelligence** — 2 modules, 8 tables
-  **Platform Operations** — 8 modules, 15 tables

Where the data lives

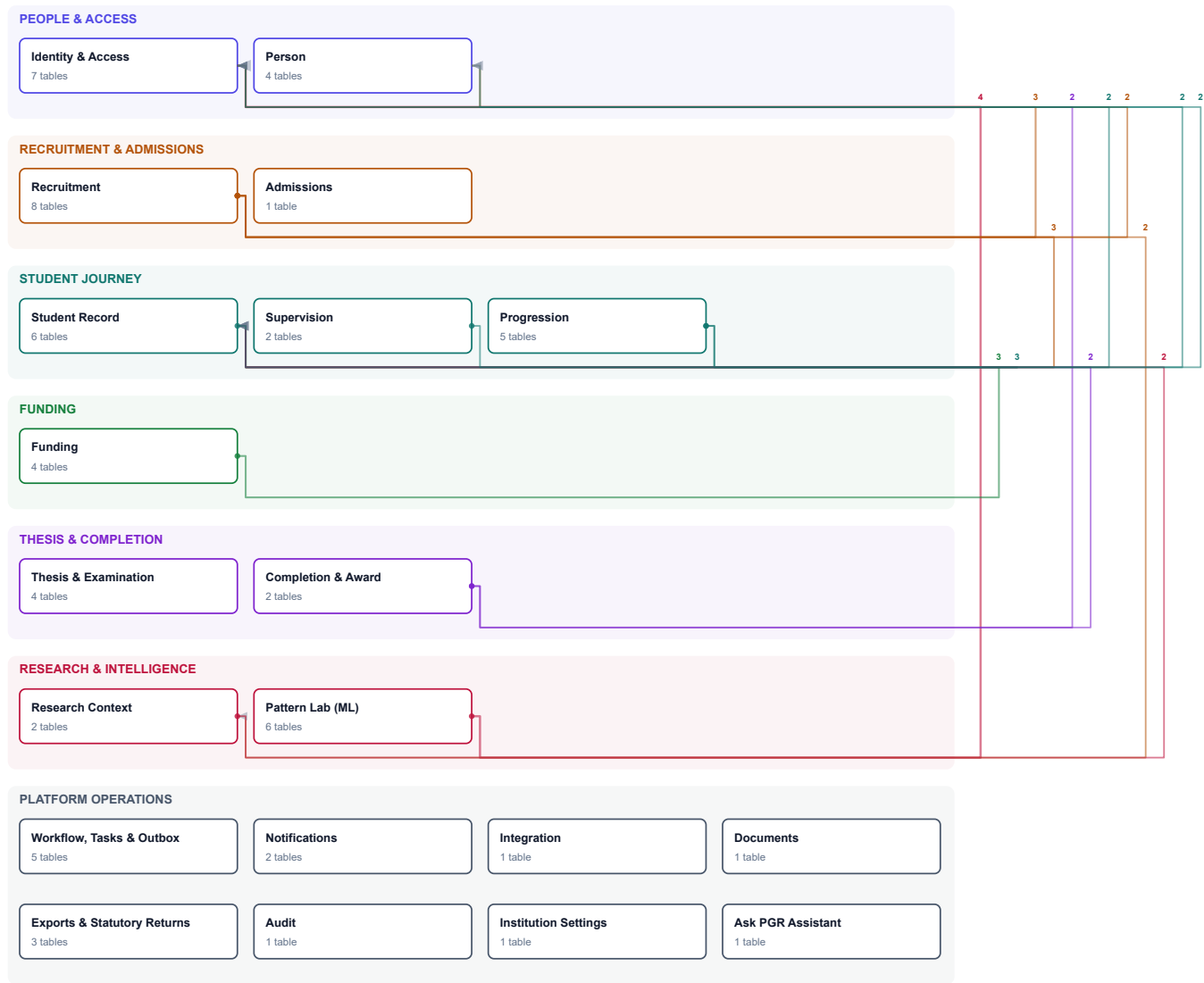
Store	Contents
PostgreSQL pgr	All 66 tables — records, workflow state, audit trail, settings, ML artifacts and predictions.
File volume STORAGE_ROOT	Uploaded document bytes only. Their metadata is the document table.

There is no second database, no cache tier and no object store. One database backup plus one volume backup captures the entire platform state.

2 · The big picture — all 66 tables by feature area

39 routes · 61 boundary-crossing keys

Each box is a feature module and how many tables it owns. A line is a **route**: one or more foreign keys pointing from the module on the left of the line to the module the arrow enters. Notice where the arrows converge — **Person**, **Student Record** and **Identity & Access** are the spine every other feature hangs off.



The 13 busiest routes are drawn; the number on each lane is how many foreign keys it carries. All 39 routes (61 keys) are tabulated later.

3 · Table inventory

every table, its owning feature and size

PEOPLE & ACCESS

Table	Cols	FKs
password_reset_token	6	1
permission	5	—
refresh_token	6	1
role	5	—
role_permission	2	2
user_role	2	2
users	10	1
person	11	—
person_contact	9	1
person_merge_record	8	1
person_relationship	8	1

RECRUITMENT & ADMISSIONS

Table	Cols	FKs
application	13	3
application_assessment	9	2
candidate_stage_history	9	2
interview	9	1
interview_panellist	6	2
offer_condition	9	1
reference_request	13	1
research_opportunity	17	5
offer	9	1

STUDENT JOURNEY

Table	Cols	FKs
department	5	—
programme	6	1
research_area	6	1
research_project	11	4
student	13	4
student_lifecycle_event	18	3
supervision_meeting	14	3
supervisor_relationship	11	2
milestone	7	2
milestone_definition	11	1
progression_appeal	11	3
progression_review	15	2
review_panel_member	7	2

FUNDING

Table	Cols	FKs
fee_waiver	13	3
funding_arrangement	17	3
funding_source	5	—
stipend_payment	13	2

THESIS & COMPLETION

Table	Cols	FKs
examination	10	1
examiner_nomination	11	3
thesis	9	1
thesis_correction	9	2
award	13	4
completion	8	1

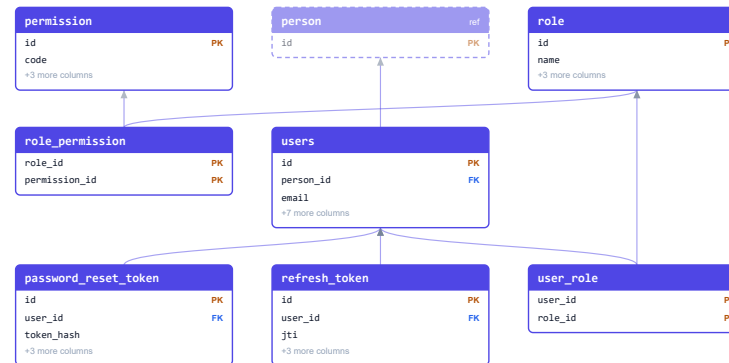
RESEARCH & INTELLIGENCE

Table	Cols	FKs
research_award	15	2
research_demand	12	4
ml_dataset	14	1
ml_finding	11	1
ml_model	7	1
ml_model_version	18	3
ml_prediction	11	3
ml_training_run	10	3

PLATFORM OPERATIONS

Table	Cols	FKs
notification	8	1
outbox_event	11	—
task	15	1
workflow_definition	10	—
workflow_instance	8	1
email_bounce	8	1
notification_preference	9	1
integration_log	10	—
document	13	—
export_job	9	—
report_field_mapping	11	1
report_profile	12	1
audit_log	11	—
institution_setting	6	1
assistant_write_intent	11	1

Who may sign in and what they may do. Roles carry permissions; users hold roles. Tokens are rows so a session can be revoked centrally.



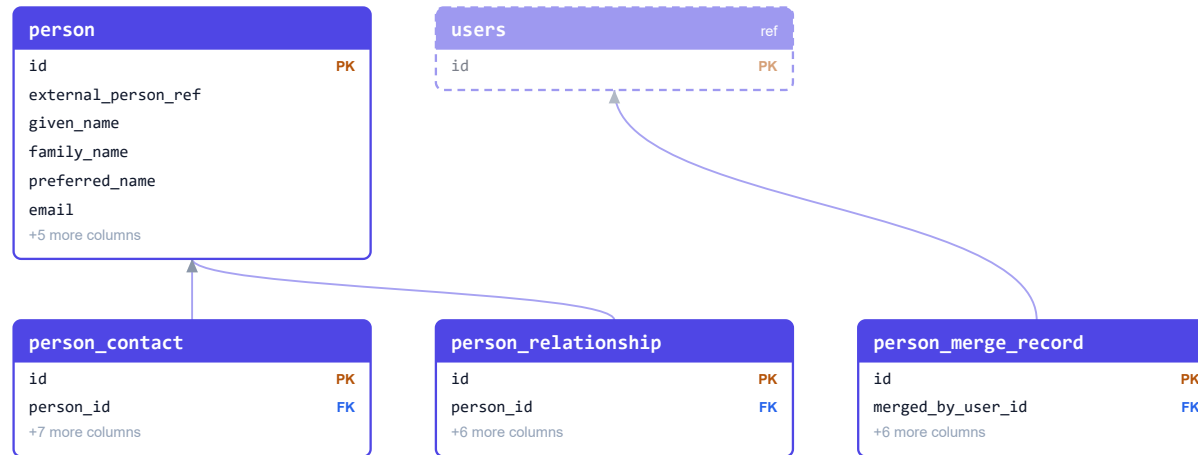
Depends on (outbound FKs)

From column	References	On delete
users.person_id	person.id	SET NULL

Referenced by (inbound FKs)

Table	Column	On delete
application_assessment	assessor_user_id → users	—
assistant_write_intent	proposed_by_user_id → users	SET NULL
award	confirmed_by_user_id → users	SET NULL
award	proposed_by_user_id → users	SET NULL
candidate_stage_history	moved_by_user_id → users	—
examiner_nomination	approved_by_user_id → users	—
fee_waiver	approved_by_user_id → users	—
institution_setting	updated_by_user_id → users	SET NULL
ml_dataset	created_by_user_id → users	SET NULL
ml_model	created_by_user_id → users	SET NULL
ml_model_version	decided_by_user_id → users	SET NULL
ml_training_run	started_by_user_id → users	SET NULL
notification	recipient_user_id → users	CASCADE
notification_preference	user_id → users	CASCADE
+10 more inbound references		

The single human record. One person_id survives every role change — applicant, student, employee, alumnus — which is why concurrent identities are rows in person_relationship rather than flags on a student.



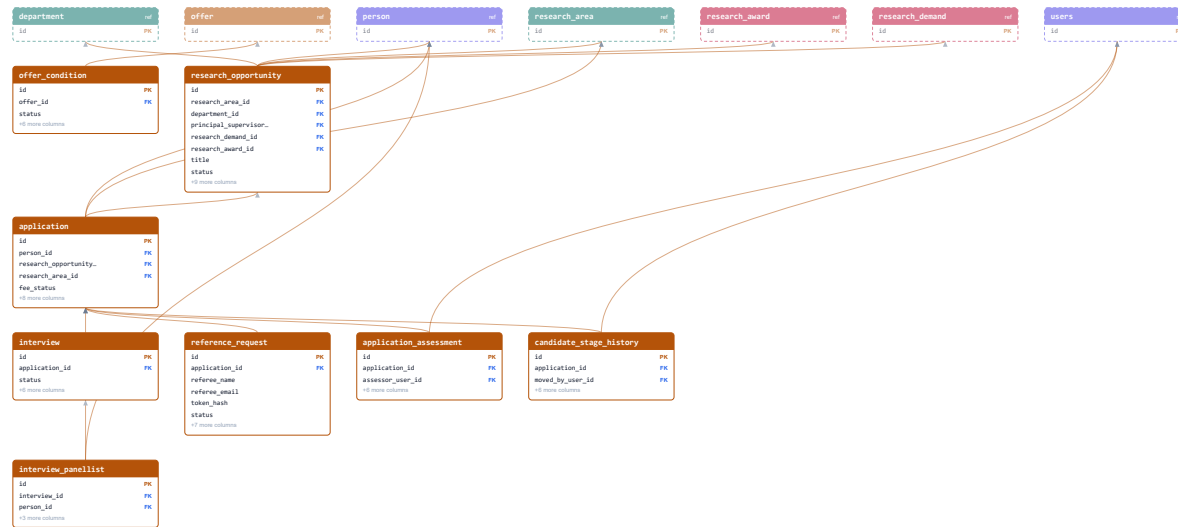
Depends on (outbound FKs)

From column	References	On delete
person_merge_record.merged_by_user_id	users.id	SET NULL

Referenced by (inbound FKs)

Table	Column	On delete
application	person_id → person	—
email_bounce	person_id → person	SET NULL
examiner_nomination	examiner_person_id → person	—
interview_panellist	person_id → person	—
research_award	principal_investigator_id → person	—
research_opportunity	principal_supervisor_id → person	—
review_panel_member	person_id → person	—
student	person_id → person	—
supervision_meeting	supervisor_person_id → person	—
supervisor_relationship	supervisor_person_id → person	—
users	person_id → person	SET NULL

From an advertised position to an assessed candidate: opportunities, applications, the stage history that proves how a decision was reached, interviews and references.



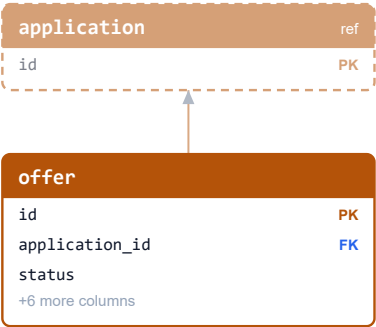
Depends on (outbound FKs)

From column	References	On delete
application.person_id	person.id	—
application.research_area_id	research_area.id	—
application_assessment.assessor_user_id	users.id	—
candidate_stage_history.moved_by_user_id	users.id	—
interview_panellist.person_id	person.id	—
offer_condition.offer_id	offer.id	CASCADE
research_opportunity.department_id	department.id	—
research_opportunity.principal_supervisor_id	person.id	—
research_opportunity.research_area_id	research_area.id	—
research_opportunity.research_award_id	research_award.id	SET NULL
research_opportunity.research_demand_id	research_demand.id	SET NULL

Referenced by (inbound FKs)

Table	Column	On delete
offer	application_id → application	CASCADE
research_project	research_opportunity_id → research_opportunity	SET NULL

The offer and its conditions. Accepting an offer creates the student record against the same person_id.



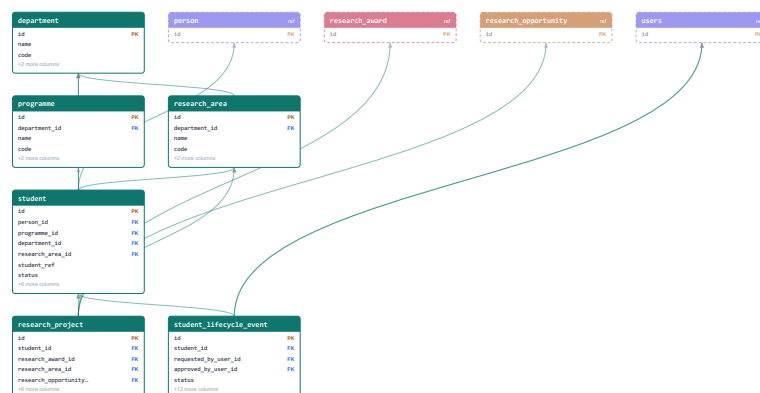
Depends on (outbound FKs)

From column	References	On delete
offer.application_id	application.id	CASCADE

Referenced by (inbound FKs)

Table	Column	On delete
offer_condition	offer_id → offer	CASCADE

The registered researcher and the reference data around them, plus the lifecycle events (suspension, extension, mode change) that legitimately move an end date.



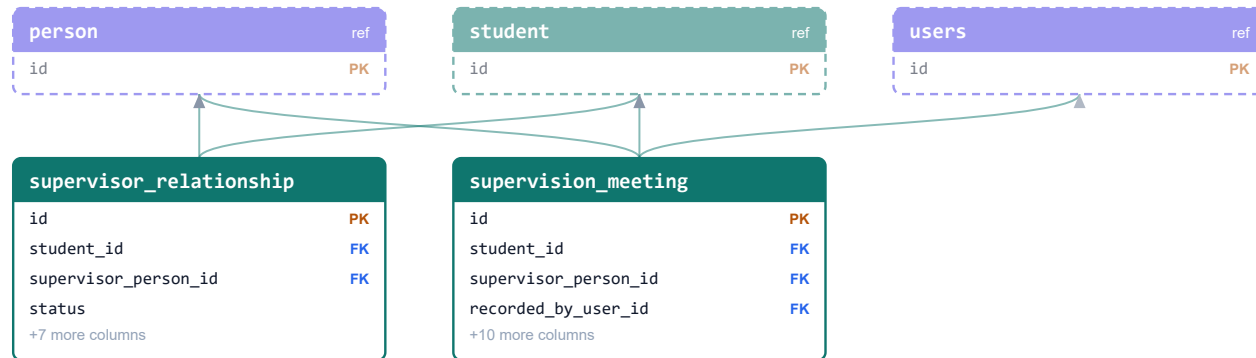
Depends on (outbound FKs)

From column	References	On delete
research_project.research_award_id	research_award.id	SET NULL
research_project.research_opportunity_id	research_opportunity.id	SET NULL
student.person_id	person.id	—
student_lifecycle_event.approved_by_user_id	users.id	—
student_lifecycle_event.requested_by_user_id	users.id	—

Referenced by (inbound FKs)

Table	Column	On delete
application	research_area_id → research_area	—
award	student_id → student	CASCADE
completion	student_id → student	CASCADE
fee_waiver	student_id → student	CASCADE
funding_arrangement	student_id → student	CASCADE
milestone	student_id → student	CASCADE
milestone_definition	programme_id → programme	CASCADE
ml_prediction	student_id → student	CASCADE
progression_appeal	student_id → student	CASCADE
research_demand	department_id → department	—
research_demand	research_area_id → research_area	—
research_opportunity	department_id → department	—
research_opportunity	research_area_id → research_area	—
stipend_payment	student_id → student	CASCADE
+3 more inbound references		

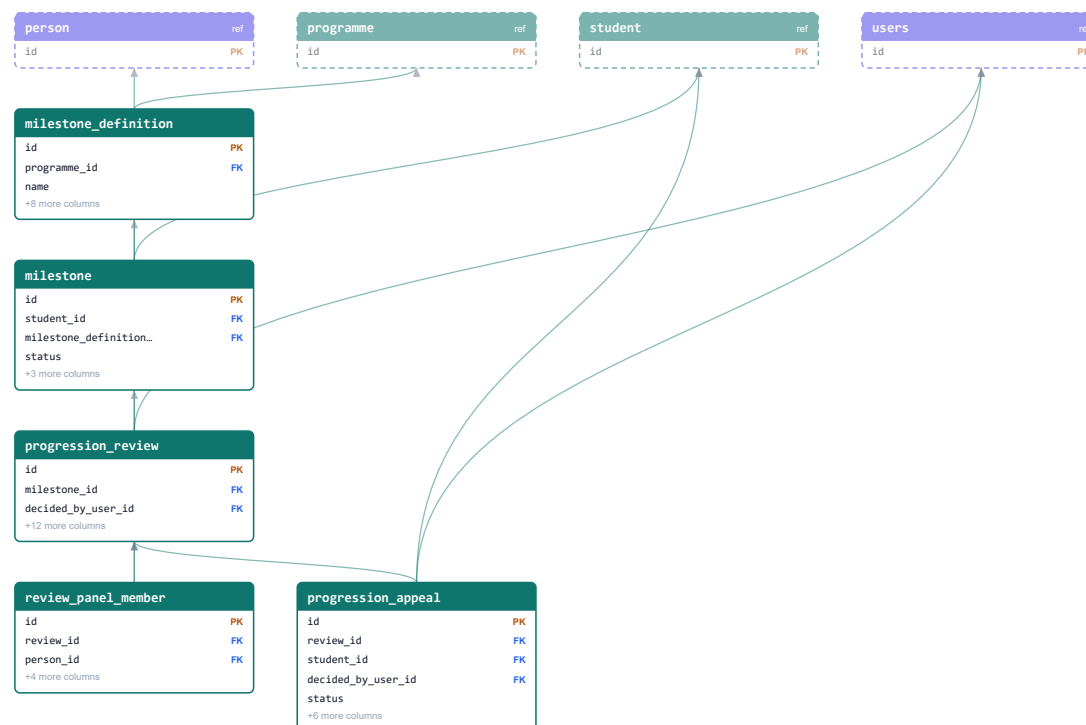
Who supervises whom, over time, and the evidence of contact. A supervisor change closes one row and opens another.



Depends on (outbound FKs)

From column	References	On delete
supervision_meeting.recorded_by_user_id	users.id	—
supervision_meeting.student_id	student.id	CASCADE
supervision_meeting.supervisor_person_id	person.id	—
supervisor_relationship.student_id	student.id	CASCADE
supervisor_relationship.supervisor_person_id	person.id	—

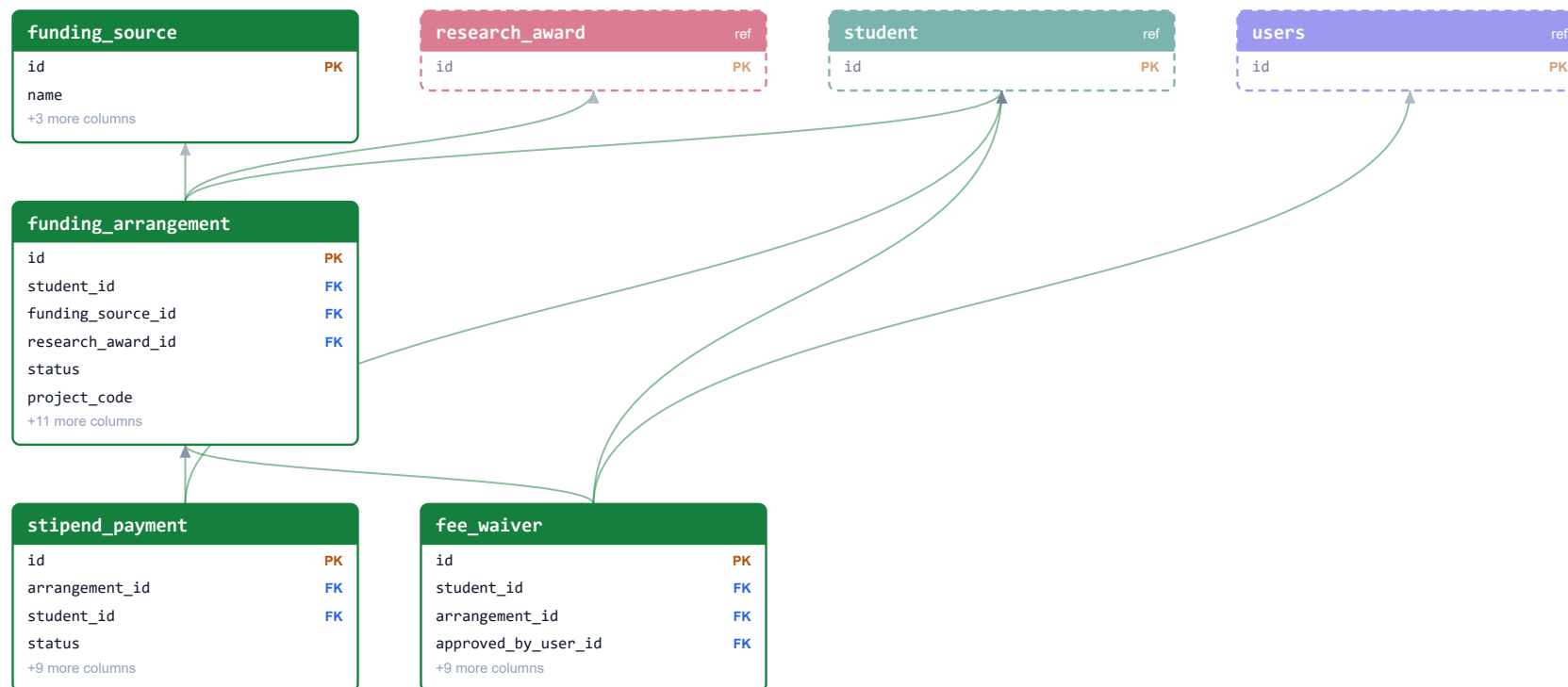
Milestones defined per programme, the student's instances of them, the review that decides them, the panel that sat, and any appeal.



Depends on (outbound FKs)

From column	References	On delete
<code>milestone.student_id</code>	<code>student.id</code>	CASCADE
<code>milestone_definition.programme_id</code>	<code>programme.id</code>	CASCADE
<code>progression_appeal.decided_by_user_id</code>	<code>users.id</code>	—
<code>progression_appeal.student_id</code>	<code>student.id</code>	CASCADE
<code>progression_review.decided_by_user_id</code>	<code>users.id</code>	—
<code>review_panel_member.person_id</code>	<code>person.id</code>	—

The money relationship — not payroll. Arrangements over time, the stipend instalment schedule, and fee waivers.



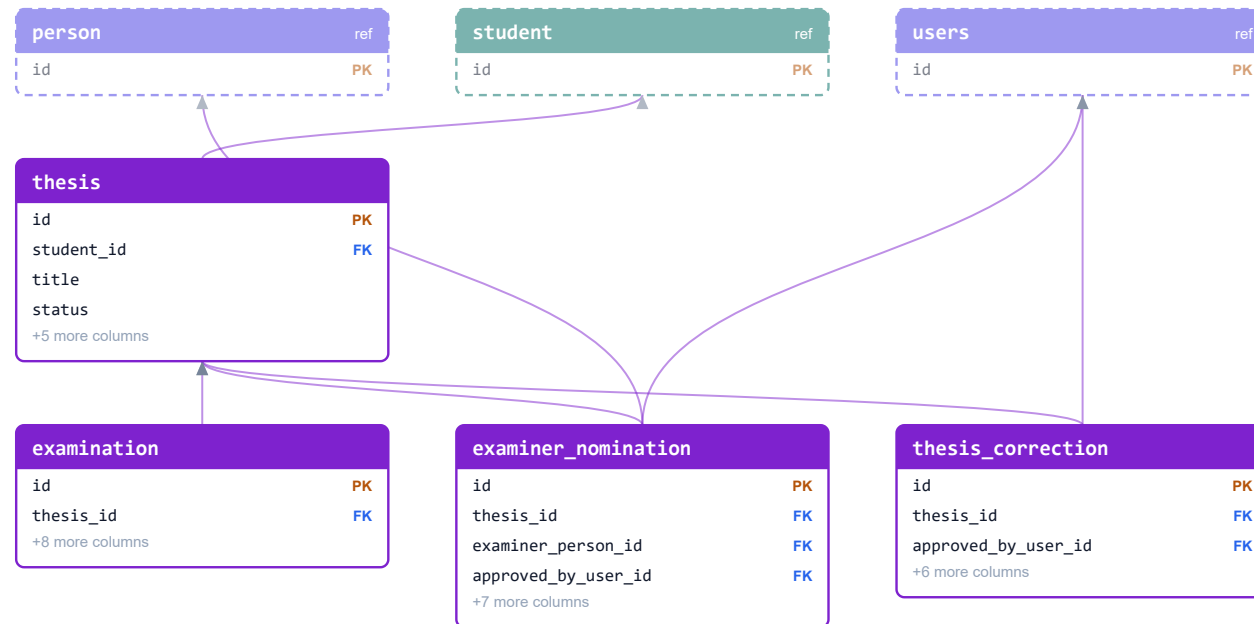
Depends on (outbound FKs)

From column	References	On delete
fee_waiver.approved_by_user_id	users.id	—
fee_waiver.student_id	student.id	CASCADE
funding_arrangement.research_award_id	research_award.id	SET NULL
funding_arrangement.student_id	student.id	CASCADE
stipend_payment.student_id	student.id	CASCADE

Referenced by (inbound FKs)

Table	Column	On delete
research_award	funder_id → funding_source	—

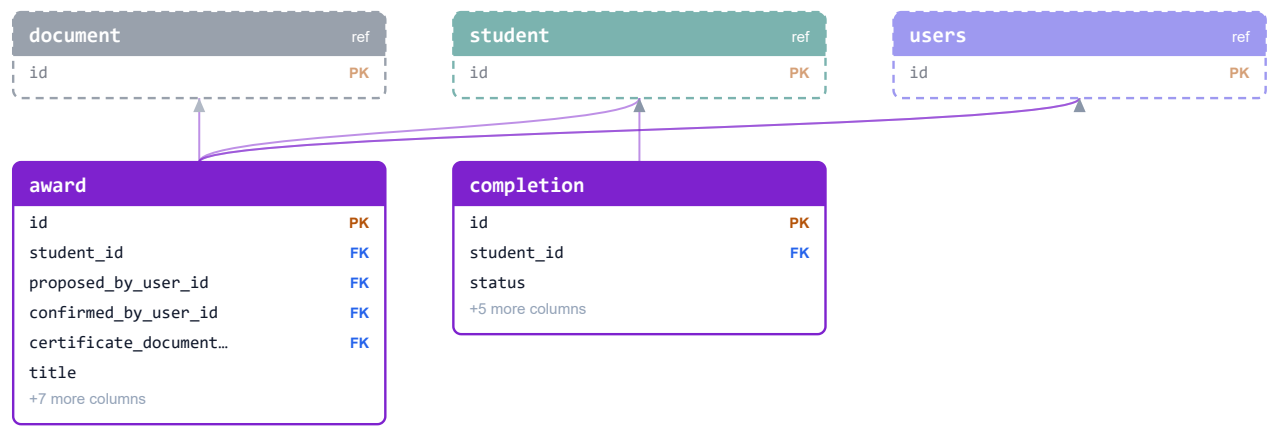
Intention to submit, the thesis, examiner nominations, the examination outcome and any corrections.



Depends on (outbound FKs)

From column	References	On delete
examiner_nomination.approved_by_user_id	users.id	—
examiner_nomination.examiner_person_id	person.id	—
thesis.student_id	student.id	CASCADE
thesis_correction.approved_by_user_id	users.id	—

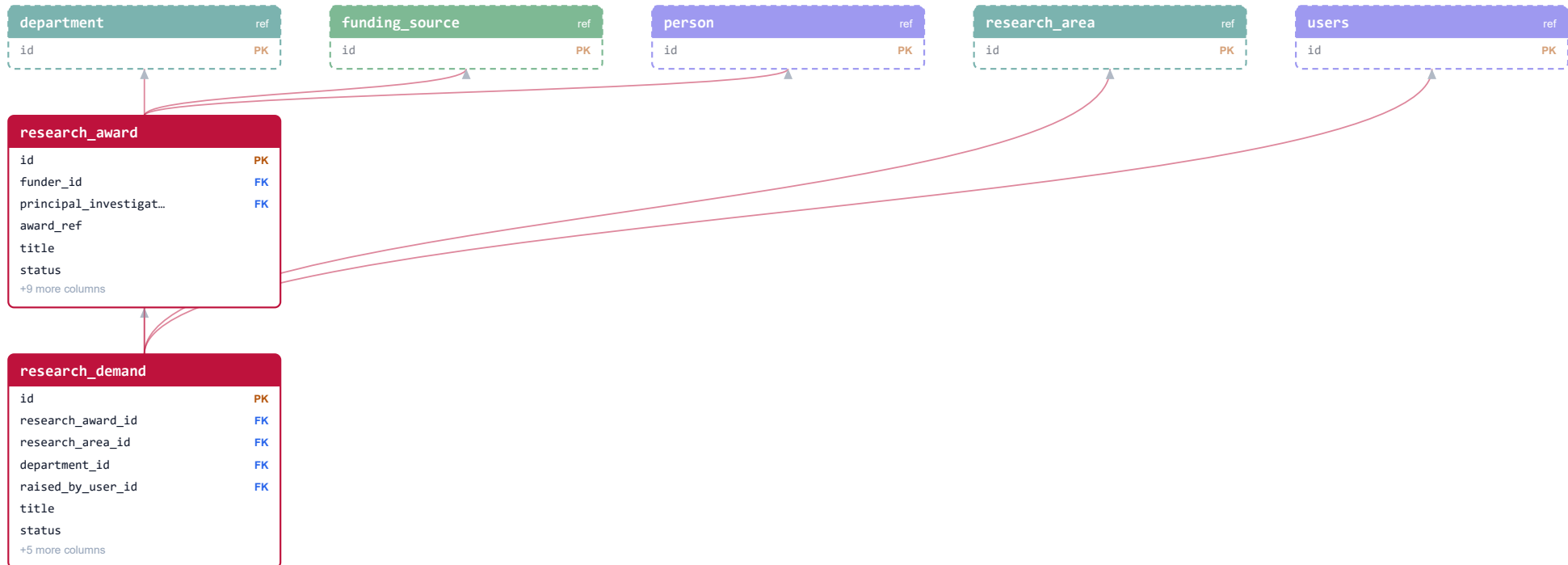
Completion confirmation and the classified award, including the published certificate.



Depends on (outbound FKs)

From column	References	On delete
award.certificate_document_id	document.id	SET NULL
award.confirmed_by_user_id	users.id	SET NULL
award.proposed_by_user_id	users.id	SET NULL
award.student_id	student.id	CASCADE
completion.student_id	student.id	CASCADE

The research context a PGR position comes from: the grant/award reference (mastered externally) and the demand raised against it.



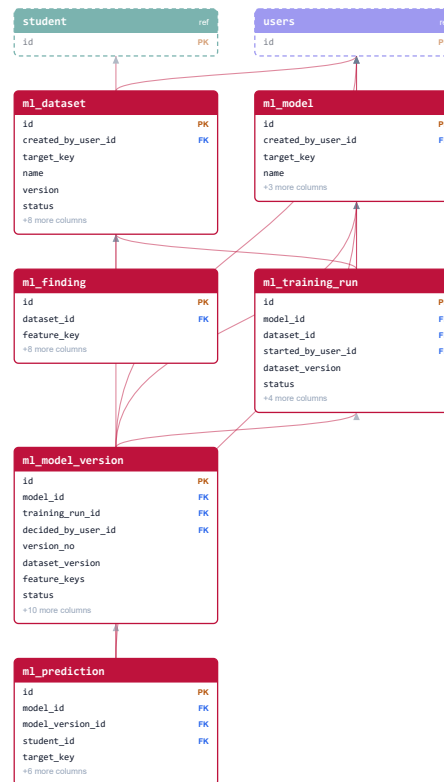
Depends on (outbound FKs)

From column	References	On delete
research_award.funder_id	funding_source.id	—
research_award.principal_investigator_id	person.id	—
research_demand.department_id	department.id	—
research_demand.raised_by_user_id	users.id	—
research_demand.research_area_id	research_area.id	—

Referenced by (inbound FKs)

Table	Column	On delete
funding_arrangement	research_award_id → research_award	SET NULL
research_opportunity	research_award_id → research_award	SET NULL
research_opportunity	research_demand_id → research_demand	SET NULL
research_project	research_award_id → research_award	SET NULL

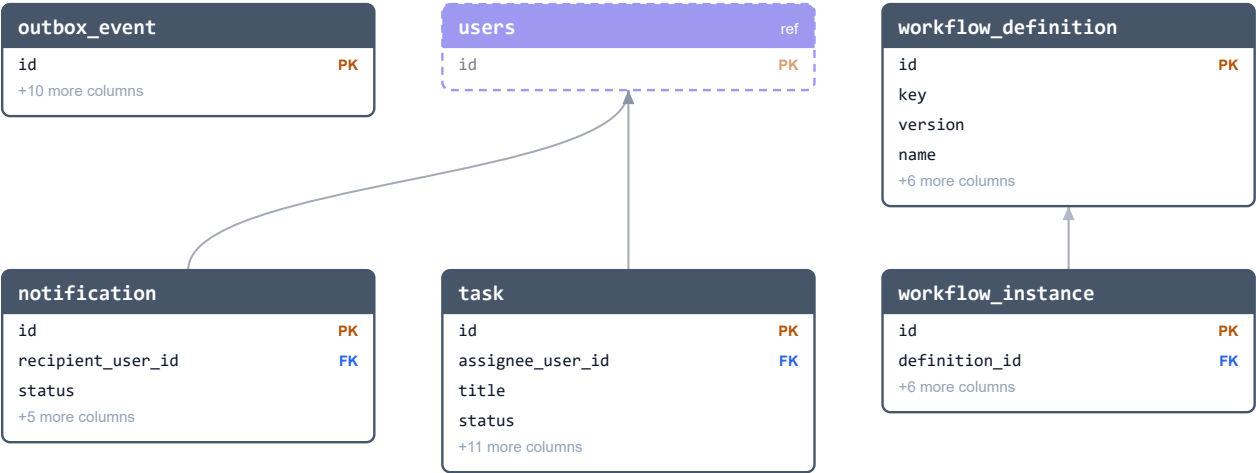
The governed ML layer. Frozen datasets, statistical findings, model versions with their artifacts and governance log, and append-only predictions.



Depends on (outbound FKs)

From column	References	On delete
ml_dataset.created_by_user_id	users.id	SET NULL
ml_model.created_by_user_id	users.id	SET NULL
ml_model_version.decided_by_user_id	users.id	SET NULL
ml_prediction.student_id	student.id	CASCADE
ml_training_run.started_by_user_id	users.id	SET NULL

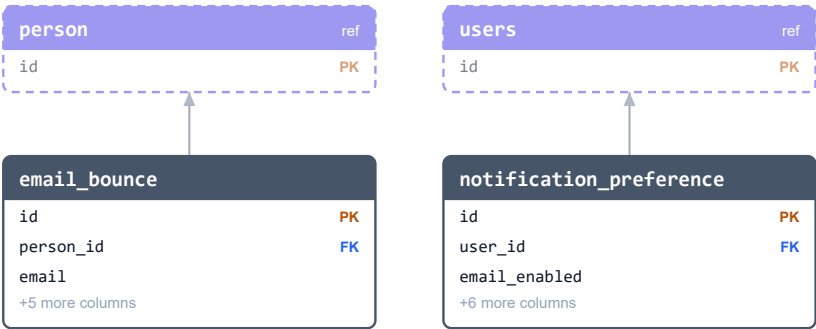
The engine behind every queue: workflow definitions and instances, human tasks, notifications, and the transactional outbox that guarantees an external message is sent exactly once.



Depends on (outbound FKs)

From column	References	On delete
notification.recipient_user_id	users.id	CASCADE
task.assignee_user_id	users.id	SET NULL

Delivery preferences per user and hard-bounce handling.



Depends on (outbound FKs)

From column	References	On delete
email_bounce.person_id	person.id	SET NULL
notification_preference.user_id	users.id	CASCADE

The message log for every inbound and outbound partner exchange.

integration_log	
id	PK
status	
+8 more columns	

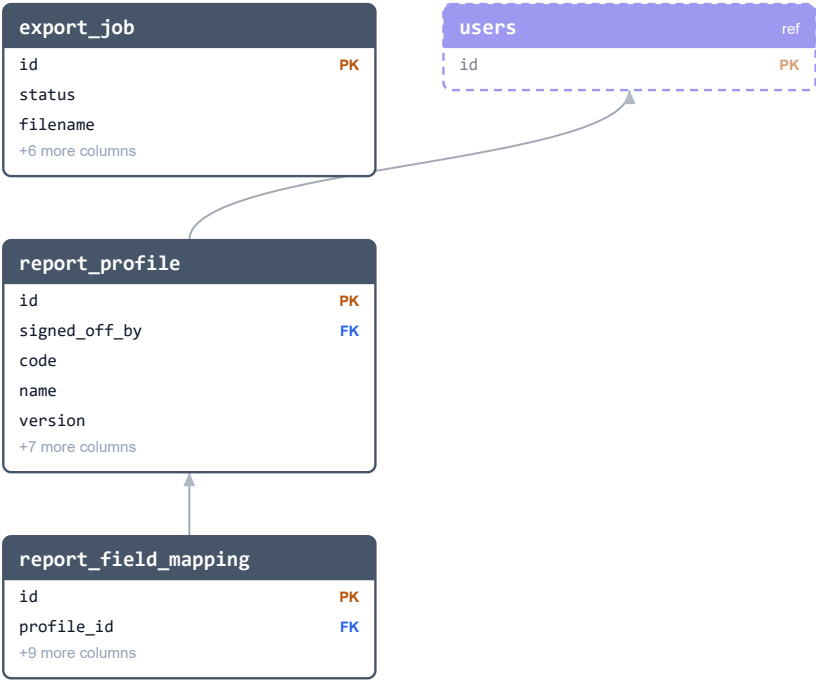
Metadata for uploaded files; the bytes live on the storage volume.

document	
id	PK
filename	
storage_key	
scan_status	
+9 more columns	

Referenced by (inbound FKs)

Table	Column	On delete
award	certificate_document_id → document	SET NULL

Export jobs plus the statutory return profiles and their field mappings — returns are configuration, not code.



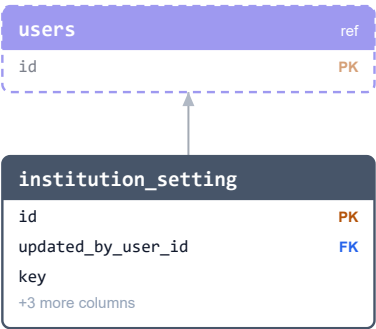
Depends on (outbound FKs)

From column	References	On delete
report_profile.signed_off_by	users.id	SET NULL

Append-only record of who did what, written by middleware.

audit_log	
id	PK
actor_email	
status_code	
+8 more columns	

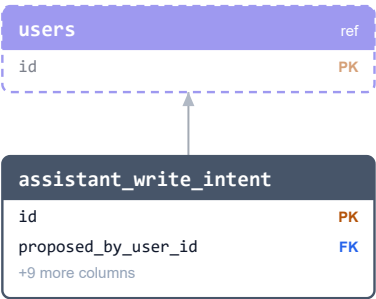
Only the institution's overridden values; defaults live in code.



Depends on (outbound FKs)

From column	References	On delete
institution_setting.updated_by_user_id	users.id	SET NULL

Proposed assistant writes awaiting human confirmation.



Depends on (outbound FKs)

From column	References	On delete
assistant_write_intent.proposed_by_user_id	users.id	SET NULL

24 · Key flows across the schema

how the tables are used in sequence

The applicant-to-alumnus lifecycle



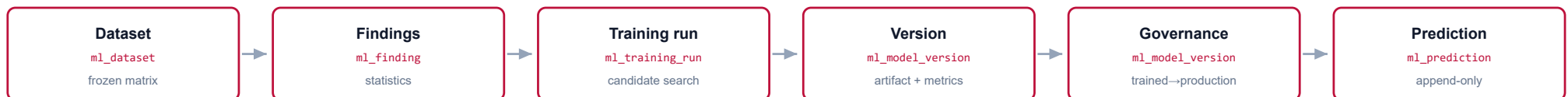
One person .id is carried through every stage above — the applicant, the student and the alumnus are the same row. Nothing is re-keyed at any handover.

Funding lineage — the chain the integrity engine walks



Every link is checked: a gap, an overlap, funding that outlives its award or a stipend exceeding the award value each raise a named finding.

Pattern Lab — dataset to governed prediction



Every prediction traces back through version → run → dataset version. Only a version in status "production" may score, and promotion needs a second human.

25 · Cross-feature relationships

all 61 boundary-crossing foreign keys

From	To	FK	Columns
Pattern Lab (ML)	Identity & Access	4	ml_dataset.created_by_user_id, ml_model.created_by_user_id +2
Funding	Student Record	3	fee_waiver.student_id, funding_arrangement.student_id +1
Progression	Student Record	3	milestone.student_id, milestone_definition.programme_id +1
Recruitment	Person	3	application.person_id, interview_panellist.person_id +1
Recruitment	Student Record	3	application.research_area_id, research_opportunity.research_area_id +1
Completion & Award	Identity & Access	2	award.proposed_by_user_id, award.confirmed_by_user_id
Completion & Award	Student Record	2	award.student_id, completion.student_id
Progression	Identity & Access	2	progression_appeal.decided_by_user_id, progression_review.decided_by_user_id
Recruitment	Identity & Access	2	application_assessment.assessor_user_id, candidate_stage_history.moved_by_user_id
Recruitment	Research Context	2	research_opportunity.research_demand_id, research_opportunity.research_award_id
Research Context	Student Record	2	research_demand.research_area_id, research_demand.department_id
Student Record	Identity & Access	2	student_lifecycle_event.requested_by_user_id, student_lifecycle_event.approved_by_user_id
Supervision	Person	2	supervision_meeting.supervisor_person_id, supervisor_relationship.supervisor_person_id
Supervision	Student Record	2	supervision_meeting.student_id, supervisor_relationship.student_id
Thesis & Examination	Identity & Access	2	examiner_nomination.approved_by_user_id, thesis_correction.approved_by_user_id
Workflow, Tasks & Outbox	Identity & Access	2	notification.recipient_user_id, task.assignee_user_id
Admissions	Recruitment	1	offer.application_id
Ask PGR Assistant	Identity & Access	1	assistant_write_intent.proposed_by_user_id
Completion & Award	Documents	1	award.certificate_document_id
Exports & Statutory Returns	Identity & Access	1	report_profile.signed_off_by

From	To	FK	Columns
Funding	Identity & Access	1	fee_waiver.approved_by_user_id
Funding	Research Context	1	funding_arrangement.research_award_id
Identity & Access	Person	1	users.person_id
Notifications	Identity & Access	1	notification_preference.user_id
Notifications	Person	1	email_bounce.person_id
Pattern Lab (ML)	Student Record	1	ml_prediction.student_id
Person	Identity & Access	1	person_merge_record.merged_by_user_id
Progression	Person	1	review_panel_member.person_id
Recruitment	Admissions	1	offer_condition.offer_id
Research Context	Funding	1	research_award.funder_id
Research Context	Identity & Access	1	research_demand.raised_by_user_id
Research Context	Person	1	research_award.principal_investigator_id
Institution Settings	Identity & Access	1	institution_setting.updated_by_user_id
Student Record	Person	1	student.person_id
Student Record	Recruitment	1	research_project.research_opportunity_id
Student Record	Research Context	1	research_project.research_award_id
Supervision	Identity & Access	1	supervision_meeting.recorded_by_user_id
Thesis & Examination	Person	1	examiner_nomination.examiner_person_id
Thesis & Examination	Student Record	1	thesis.student_id

Deletion policy

Hard DELETE is reachable through the API in five places only. Everything else transitions, closes or is pseudonymised.

What can be deleted	Why it is safe
person_contact	A wrong phone number is not history.
document	Uploads can be mistakes; the audit log records the act.
report_field_mapping	Draft configuration — refused once the profile is signed off.
Reference values department, research_area, programme, funding_source	Refused while any row still references them; the error names what is using it.
institution_setting	Not data loss — deleting the override restores the shipped default.

Cascade behaviour

ON DELETE	FKs	Meaning
CASCADE	45	Child rows belong to the parent and go with it (e.g. a student's milestones).
no action	37	Referenced rows cannot be removed while children exist — the database refuses, which is what protects reference data.
SET NULL	21	The reference is optional context — the child survives (e.g. who approved a row).

Identity and history

One person, many roles. person_relationship holds a row per identity with its own validity window, so a research assistant who is also a PGR student is one person with two open relationships — not two records to reconcile.

Merging is recorded. person_merge_record keeps the losing id and the rewritten references, so a merge can be explained afterwards.

Erasure is pseudonymisation. A GDPR erase blanks identifying columns and keeps the structural rows, so statutory counts stay correct while the individual becomes unidentifiable.

Guarantees that depend on specific tables

Table	Guarantee
outbox_event	An external message is written in the same transaction as the change that caused it — no lost or phantom notifications.
integration_log	Inbound messages are deduplicated by (system, source_id): a partner may safely retry.
audit_log	Append-only; written by middleware rather than by each endpoint, so nothing can quietly skip it.
student_lifecycle_event	Holds the days applied by each suspension or extension, so an expected end date is always reconstructable.
ml_model_version	Carries the governance log and the trained artifact together — a prediction can always be traced to an approved version.
ml_prediction	Append-only, so monitoring can compare an old prediction against what actually happened.